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CROSS-VALIDATION, INFLUENTIAL OBSERVATIONS AND SELECTION OF VARIABLES IN CHEMOMETRIC STUDIES OF WINES BY PRINCIPAL COMPONENTS ANALYSIS†

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SUMMARY

Cross-validatory estimation of the bilinear model based on principal components is reviewed and Krzanowski's modification of Wold's procedure is described. Two different types of residuals useful for checking model adequacy are defined and indices measuring the influence of each observed unit on the estimates of the parameters are discussed. A method for the selection of variables derived from Procrustes analysis is described. Results arising from the study of two sets of enological data are given.

KEY WORDS Principal component Cross-validation Selection of variables Residual Influential observation Procrustes analysis Chemometrics Wine analysis

INTRODUCTION

Modern instrumental methods of chemical analysis of foods enable the concentration of a very high number of components to be determined accurately and simultaneously. Thus an extensive chemical characterization of samples takes, at present, a matter of hours. Frequently, however, as a consequence of a thorough survey, redundant variables are included in the observed data and useful information can be partly obscured by data variability and distorted by outliers, so that statistical methods are required to retain 'structure' and discard 'noise'.

Customary problems to be solved at the very beginning of an investigation are (i) the recognition of influential observations, (ii) the determination of the data dimensionality and (iii) the identification of 'important' variables containing most of the information supplied by the data. Influential observations, by definition, cause considerable changes in the results of a statistical analysis when they are discarded from the data. Influence analysis[1] can also be of use in recognizing untypical cases and/or outliers. Depending on the structure of the data, the assessment of the dimensionality is carried out by principal components analysis (PCA) or

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other factorial methods such as canonical variates analysis and canonical correlation analysis. However, in PCA, as for the other methods, each component is a linear transformation of all the original variables, so the reduction of dimensionality obtained by retaining only a subset of the principal components does not imply a corresponding reduction in the number of variables to be observed. The selection of variables has received great attention in multiple regression[2] and discriminant analysis;[3] in PCA a few methods for the selection of optimal subsets of variables have been proposed by Jolliffe,[4,5] McCabe[6] and more recently, by Krzanowski.[7]

In this work PCA is used as a general technique for studying enological data obtained by chemical analysis. Known results concerning least squared estimation of the bilinear model based on principal components are recalled. Important topics include cross-validatory estimation of the number of components, detection of influential observations and selection of variables. We discuss theoretical and practical issues arising from the study of (a) a multi-group data set including concentration measurements of 18 polyphenol fractions on 22 samples of three Italian red wines, Chianti Classico, Brunello di Montalcino and Marzemino del Trentino (Polyphenol data), and (b) a single-group data set including 19 classical chemical determinations and the concentration measurements of nine aroma compounds on 34 samples of the Italian white wine Prosecco di Conegliano–Valdobbiadene (Prosecco data).

STATISTICS

A model for data in PCA

Least squares estimation of the bilinear model

In general, the initial data for a chemometric investigation of enological products are the values of p numerical variables (chemical measurements) observed on n samples. These data are usually arranged in a matrix $\mathbf{X}(n \times p)$ whose column vectors $\mathbf{x}^{(j)}$, $j = 1, \ldots, p$, contain, for each variable, the observed values on the n samples while the row vectors $\mathbf{x}_i^{\mathrm{T}}$, $i = 1, \ldots, n$, record, for each sample, the scores on the p variables. The value of the jth variable on the ith sample is denoted by x_{ij}, $i = 1, \ldots, n$, $j = 1, \ldots, p$.

In PCA we assume that the observed data are generated by the model[8,9]

$$\mathbf{X} = \mathbf{1}\mathbf{a}^{\mathrm{T}} + \mathbf{Z}\mathbf{B}^{\mathrm{T}} + \mathbf{E} \tag{1}$$

where $\mathbf{1}$ is the $n \times 1$ vector with all the components equal to one, $\mathbf{a} = (a_1, \ldots, a_p)^{\mathrm{T}}$, $\mathbf{Z}$ is an $n \times q$ matrix whose columns are determinations of $q < p$ uncorrelated underlying factors, $\mathbf{B}$ is a $p \times q$ matrix of loadings and $\mathbf{E}$ is an $n \times p$ matrix of residuals. An important feature of the bilinear model (1) is that in the usual practical situation the number of factors is unknown and must be estimated.

Let $\tilde{\mathbf{X}}$ be the mean-centred data matrix and let p be the rank of $\tilde{\mathbf{X}}$. From the singular value decomposition (SVD) theorem[10] we can write

$$\tilde{\mathbf{X}} = \mathbf{V}\mathbf{D}\mathbf{U}^{\mathrm{T}}$$

where $\mathbf{V}(n \times p)$ has orthonormal column vectors, $\mathbf{U}(p \times p)$ is an orthogonal matrix and $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_p)$ with $d_1 \geqslant \ldots \geqslant d_p > 0$. The principal components of $\tilde{\mathbf{X}}$ are the column vectors of the $n \times p$ matrix $\mathbf{C} = \tilde{\mathbf{X}}\mathbf{U} = \mathbf{V}\mathbf{D}$. For any given q, $1 \leqslant q < p$, let the matrices $\mathbf{U}$, $\mathbf{V}$

and $\mathbf{D}$ be partitioned as $\mathbf{U} = (\mathbf{U}_1 : \mathbf{U}_2)$, $\mathbf{V} = (\mathbf{V}_1 : \mathbf{V}_2)$ and

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_1 & 0 \\ 0 & \mathbf{D}_2 \end{bmatrix}$$

where $\mathbf{U}_1$ and $\mathbf{V}_1$ are $p \times q$ and $n \times q$ respectively and $\mathbf{D}_1 = \mathrm{diag}(d_1, \ldots, d_q)$. Then the column vectors of the $n \times q$ matrix $\mathbf{C}_1 = \tilde{\mathbf{X}}\mathbf{U}_1 = \mathbf{V}_1\mathbf{D}_1$ are the first q principal components of $\tilde{\mathbf{X}}$.

The least squares estimation of $\mathbf{a}$, $\mathbf{Z}$ and $\mathbf{B}$ in equation (1), conditional on a fixed value of q, is a two-step procedure.[9] First, the estimate of $\mathbf{a}$ turns out to be, for each q, the vector $\bar{\mathbf{x}} = (\bar{x}_1, \ldots, \bar{x}_p)^{\mathrm{T}}$ of the sample means. Secondly, conditional on q, the estimate of $\mathbf{Z}$ is the matrix $\mathbf{C}_1$ and the estimate of $\mathbf{B}$ is the matrix $\mathbf{U}_1$. Thus to the $\mathbf{X}$ matrix is associated the predictor

$$\mathbf{X}^{(q)} = \mathbf{1}\bar{\mathbf{x}}^{\mathrm{T}} + \mathbf{C}_1\mathbf{U}_1^{\mathrm{T}}$$

where, for $i = 1, \ldots, n$ and $j = 1, \ldots, p$, x_{ij} is estimated by

$$x_{ij}^{(q)} = \bar{x}_j + \sum_{l=1}^{q} c_{il}u_{jl} = \bar{x}_j + \sum_{l=1}^{q} d_l v_{il} u_{jl}$$

The difference $\mathbf{E}^{(q)} = \mathbf{X} - \mathbf{X}^{(q)}$ is the $n \times p$ matrix of the estimated residuals. Thus to each sample there correspond the vector $\mathbf{x}_i$ of the observed data, the vector $\mathbf{x}_i^{(q)}$ of the predicted values, which is best calculated as

$$\mathbf{x}_i^{(q)} = \bar{\mathbf{x}} + \mathbf{U}_1\mathbf{U}_1^{\mathrm{T}}(\mathbf{x}_i - \bar{\mathbf{x}})$$

and the vector $\mathbf{e}_i^{(q)}$ of the residuals, given by

$$\mathbf{e}_i^{(q)} = \mathbf{x}_i - \mathbf{x}_i^{(q)} = (\mathbf{I}_p - \mathbf{U}_1\mathbf{U}_1^{\mathrm{T}})(\mathbf{x}_i - \bar{\mathbf{x}}) \tag{2}$$

where $\mathbf{I}_p$ is the identity matrix of order p.

Cross-validatory estimation of the number of components

When q varies in the set $\{1, 2, \ldots, p-1\}$, equation (1) describes a family of statistical models which differ in the number of underlying factors, i.e. in the dimensionality of the structural component of the process which is supposed to generate the observations. Thus we have to choose the 'best' model in the light of the data. To this end we adopt the predictive approach, or cross-validation, first proposed in the PCA context by Wold[11] and later improved by Eastment and Krzanowski.[12]

The cross-validatory estimation of q requires the partition of the set $\mathscr{X}$ of the n samples into G partial subsets $\mathscr{X}_1, \ldots, \mathscr{X}_g$. For $g = 1, \ldots, G$, the observations in $\mathscr{X}_g$ are discarded from $\mathscr{X}$ and the reduced data set $\mathscr{X}_{(-g)}$ is used to estimate the parameters of the model described by equation (1) with $q = q_0$, $1 \leqslant q_0 \leqslant p-1$. By these estimates, predicted values for the discarded data belonging to $\mathscr{X}_g$ can be calculated and the discrepancy between actual and predicted values can be evaluated by a suitable function PRESS(q_0) (prediction sum of squares). The whole procedure is repeated for each value of q, thus obtaining the corresponding value PRESS(q). The estimate of q is then calculated as a function of PRESS(1), ..., PRESS($p-1$).

The basic principle in the above procedure is that the same data must not be used both in the estimation step and in the step of assessment of the estimated model. Also, the size of the deleted subsets should be as small as possible to maximize the efficiency of the estimates. To comply with such requirements, Eastment and Krzanowski proposed the following deletion scheme in the Wold procedure. For $i = 1, \ldots, n$, let $\mathbf{X}_{(i)}$ be the matrix $\mathbf{X}$ with the ith row deleted and let $\tilde{\mathbf{X}}_{(i)}$ be the corresponding mean-centred data matrix. For $j = 1, \ldots, p$, let $\tilde{\mathbf{X}}^{(j)}$ be the

matrix $\tilde{\mathbf{X}}$ with the jth column deleted. From the SVD theorem

$$\tilde{\mathbf{X}}_{(i)} = \bar{\mathbf{V}}\bar{\mathbf{D}}\bar{\mathbf{U}}^{\mathrm{T}}$$

where $\bar{\mathbf{V}} = (\bar{v}_{st})$, $\bar{\mathbf{D}} = \mathrm{diag}(\bar{d}_1, \ldots, \bar{d}_p)$ and $\bar{\mathbf{U}} = (\bar{u}_{st})$. Similarly

$$\tilde{\mathbf{X}}^{(j)} = \hat{\mathbf{V}}\hat{\mathbf{D}}\hat{\mathbf{U}}^{\mathrm{T}}$$

where $\hat{\mathbf{V}} = (\hat{v}_{st})$, $\hat{\mathbf{D}} = \mathrm{diag}(\hat{d}_1, \ldots, \hat{d}_{p-1})$ and $\hat{\mathbf{U}} = (\hat{u}_{st})$. For $i = 1, \ldots, n$ and $j = 1, \ldots, p$, the predicted value of x_{ij} is now defined as

$$\hat{x}_{ij}^{(q)} = \bar{x}_{(i)j} + \sum_{l=1}^{q} (\hat{v}_{il}\hat{d}_l^{1/2})(\bar{u}_{jl}\bar{d}_l^{1/2}), \quad q = 1, \ldots, p-1 \tag{3}$$

where $\bar{x}_{(i)j} = (n-1)^{-1} \sum_{s=1, s \neq i}^{n} x_{sj}$ and the terms $(\hat{v}_{il}\hat{d}_l^{1/2})(\bar{u}_{jl}\bar{d}_l^{1/2})$, $l = 1, \ldots, q$, are obtained from the SVD of $\tilde{\mathbf{X}}$ after either the ith row or the jth column is discarded. Thus, for each $q = 1, \ldots, p-1$, x_{ij} is not used to determine its prediction $\hat{x}_{ij}^{(q)}$ and, at the same time, the predictor given by equation (3) makes an efficient use of the observed data.

Note that once the matrices $\mathbf{V}$, $\mathbf{D}$ and $\mathbf{U}$ in the SVD $\tilde{\mathbf{X}} = \mathbf{V}\mathbf{D}\mathbf{U}^{\mathrm{T}}$ are computed, the algorithms for the SVD update given by Bunch and Nielsen[13] and Bunch *et al.*[14] allow the matrices $\hat{\mathbf{V}}$, $\hat{\mathbf{D}}$ and $\bar{\mathbf{U}}$, $\bar{\mathbf{D}}$ to be determined extremely quickly, thus permitting a practical exploitation of Krzanowski's estimator.

An obvious measure of the discrepancy between the observed values x_{ij} and the corresponding predictions $\hat{x}_{ij}^{(q)}$ is the function

$$\mathrm{PRESS}(q) = \sum_{i=1}^{n} \sum_{j=1}^{p} (x_{ij} - \hat{x}_{ij}^{(q)})^2$$

which is defined for $q = 0, 1, \ldots, p-1$ if we put

$$\mathrm{PRESS}(0) = \sum_{i=1}^{n} \sum_{j=1}^{p} (x_{ij} - \bar{x}_{(i)j})^2$$

The selection of q can be based on the following W-ratio:

$$W(q) = \frac{[\mathrm{PRESS}(q-1) - \mathrm{PRESS}(q)]/D(q)}{\mathrm{PRESS}(q)/DR(q)}$$

where $D(q)$ are the degrees of freedom required to estimate the qth component and $DR(q)$ are the degrees of freedom remaining after the estimation of the qth component. By taking into account normalization and orthogonality constraints on the eigenvectors of the matrices $\tilde{\mathbf{X}}^{\mathrm{T}}\tilde{\mathbf{X}}$ and $\tilde{\mathbf{X}}\tilde{\mathbf{X}}^{\mathrm{T}}$, it is easy to show that $D(q) = n + p - 2q$, while $DR(q)$ can be iteratively calculated as

$$DR(q) = DR(q-1) - D(q), \quad q = 1, \ldots, p-1$$

with $DR(0) = (n-1)p$.

The statistics $W(q)$ can be thought of as the information gain supplied by the qth component divided by the information supplied, on the average, by the residual components $(q+1)$, $(q+2), \ldots$. Thus 'important' components should have $W(q)$-values higher than one. The empirical criterion suggested by Krzanowski[15] is to take as the optimal number of components the highest value of q for which $W(q) > 0{\cdot}9$.

Diagnostic statistics

In parallel with the cross-validatory estimation of the bilinear model, the following useful diagnostics are produced: (a) sums of the squares of the residuals; (b) sample influence curves

for eigenvalues and eigenvectors of the covariance matrix; (c) jack-knife estimates of the eigenvalues of the covariance matrix and of the corresponding standard deviations.

Residual analysis

As in regression analysis, examination of residuals arising in PCA is helpful for checking model adequacy and recognizing possible outliers. Let $q < p$ be the estimated number of 'significant' components. Then, for $i = 1, \ldots, n$, the vector of estimated residuals corresponding to the ith sample is given by equation (2) and a measure of fit is the sum of the squared components of this vector, namely

$$Q_i = \mathbf{e}_i^{(q)\mathrm{T}}\mathbf{e}_i^{(q)} = (\mathbf{x}_i - \bar{\mathbf{x}})^{\mathrm{T}}(\mathbf{I}_p - \mathbf{U}_1\mathbf{U}_1^{\mathrm{T}})(\mathbf{x}_i - \bar{\mathbf{x}})$$

Large values of Q_i could suggest that the ith sample may be an outlier. Formal outlier tests can be performed when prediction relies on large-sample estimates of the population mean vector and covariance matrix. In this case it has been shown[16] that Q_i is approximately normally distributed.

Let $\hat{\mathbf{x}}_i^{(q)}$ be the vector $(\hat{x}_{i1}^{(q)}, \ldots, \hat{x}_{ij}^{(q)}, \ldots, \hat{x}_{ip}^{(q)})^{\mathrm{T}}$, with $\hat{x}_{ij}^{(q)}$, $j = 1, \ldots, p$, given by equation (3). If $\mathbf{x}_i^{(q)}$ in equation (2) is replaced by its cross-validated counterpart $\hat{\mathbf{x}}_i^{(q)}$, then an alternative version of the vector of the residuals, denoted by $\hat{\mathbf{e}}_i^{(q)} = \mathbf{x}_i - \hat{\mathbf{x}}_i^{(q)}$, is also available. These predicted residuals are mostly useful when studying the properties of PRESS- and W-statistics.

Influence curves

The basic idea in influence analysis is to introduce small perturbations in the formulation of a problem and to observe how these perturbations affect the results (e.g. the value of an estimator). Particular attention is devoted to the perturbation scheme in which the sample units are discarded one at a time. Such a procedure can be useful for identifying untypical observations and revealing subsets of the observation space with inadequate stability of estimates and predictions. In this case the most important tool is the sample influence curve (SIC), which measures the effect of omitting the ith unit from the sample, $i = 1, \ldots, n$.

In PCA, SICs with respect to the estimates of eigenvalues λ_j and eigenvectors $\boldsymbol{\alpha}_j$ of the population covariance matrix have been investigated by Critchley.[17,18] Let $\mathbf{S} = n^{-1}\tilde{\mathbf{X}}^{\mathrm{T}}\tilde{\mathbf{X}}$ be the sample covariance matrix and let $\mathbf{S}_{(i)} = (n-1)^{-1}\tilde{\mathbf{X}}_{(i)}^{\mathrm{T}}\tilde{\mathbf{X}}_{(i)}$. For $j = 1, \ldots, p$, the SIC of the jth eigenvalue is defined as

$$\mathrm{SIC}_i(\lambda_j) = (n-1)(l_j - l_{(i)j}), \quad i = 1, \ldots, n$$

where l_j $(= n^{-1}d_j^2)$ is the jth highest eigenvalue of $\mathbf{S}$ and $l_{(i)j}$ $(= (n-1)^{-1}\bar{d}_j^2)$ is the corresponding eigenvalue of $\mathbf{S}_{(i)}$. In the same way the SIC of the jth eigenvector is

$$\mathrm{SIC}_i(\boldsymbol{\alpha}_j) = (n-1)(\mathbf{u}_j - \mathbf{u}_{(i)j}), \quad i = 1, \ldots, n$$

where $\mathbf{u}_j$ is the eigenvector of $\mathbf{S}$ associated with l_j and $\mathbf{u}_{(i)j}$ $(= \bar{\mathbf{u}}_j)$ is the corresponding eigenvector of $\mathbf{S}_{(i)}$. If $\mathbf{U}_k$ $(p \times k)$ is the matrix of the first k eigenvectors of $\mathbf{S}$ and $\mathbf{U}_{(i)k}$ $(p \times k)$ is the matrix of the corresponding eigenvectors of $\mathbf{S}_{(i)}$, $i = 1, \ldots, n$, then a scalar measure of the influence of the ith observation on the configuration defined by the first k principal components is given by the Euclidean norm $\| \mathbf{C}_k - \mathbf{C}_{(i)k} \|$, where $\mathbf{C}_k = \tilde{\mathbf{X}}\mathbf{U}_k$ and $\mathbf{C}_{(i)k} = (\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}_{(i)}^{\mathrm{T}})\mathbf{U}_{(i)k}$.

Since at present no distributional result is available, the SICs should be used as empirical indices describing the impact of each sample on different, though not independent, aspects of PCA. Thus the components of the p-vector $\mathrm{SIC}_i(\boldsymbol{\alpha}_j)$ measure the influence of the ith

observation on the loadings which define the jth principal component, and this information is important when studying the relationship between a particular component and the observed variables. An overall index of 'stability' of the loadings is the Euclidean norm $P_i = (n-1)\,\|\,\tilde{\mathbf{u}}_j - \tilde{\mathbf{u}}_{(i)j}\,\|$, $i = 1, \ldots, n$ where the components of the vectors $\tilde{\mathbf{u}}_j$ and $\tilde{\mathbf{u}}_{(i)j}$ are the absolute values of the corresponding components of $\mathbf{u}_j$ and $\mathbf{u}_{(i)j}$. On the other hand, $\mathrm{SIC}_i(\lambda_j)$ measures the influence on the variance of the jth component, while $\|\,\mathbf{C}_k - \mathbf{C}_{(i)k}\,\|$ accounts for the displacements of the points in the k-components configuration, caused by the omission of the ith observation.

Jack-knife eigenvalues and standard deviations

The Quenouille–Tuckey jack-knife[19] is a general method designed to reduce the bias of an estimate and to evaluate its sampling variability. In PCA the interest is focused on the variance of l_j, the estimate of λ_j. For $j = 1, \ldots, p$, the jack-knife estimate of λ_j is defined as

$$\hat{l}_j = n l_j - [(n-1)/n] \sum_{i=1}^{n} l_{(i)j}$$

and the variance of $\hat{l}_j$ (and of l_j) is estimated by

$$\hat{s}_j^2 = [n(n-1)]^{-1} \sum_{i=1}^{n} \left(l_{(i)j} - n^{-1} \sum_{i=1}^{n} l_{(i)j} \right)^2$$

Selection of variables in PCA

PCA is often used as a dimension-reducing technique, i.e. to determine $q \ll p$ new variables containing most of the information supplied by the data. However, each principal component is a linear combination of the observed variables and then, even if the data dimensionality can be reduced from p to q, all the observed variables are again necessary to derive the new ones. On the contrary, in the applications one wants to reduce not only the space dimension but also the number of variables that should be observed in the future.

Pioneering work on this subject has been done by Jolliffe[4,5] and McCabe.[6] More recently, Krzanowski[7] suggested the use of Procrustes analysis[10] to determine subsets of the observed variables reproducing as closely as possible the main features of the data reflected by the selected q principal components.

Krzanowski's method

Let us suppose that the data dimensionality is $q < p$. We denote with $\mathbf{C}_X$ the $n \times q$ matrix of the first q principal components of $\mathbf{X}$. For any given subset of k observed variables, with $q \leqslant k < p$, let $\mathbf{Y}$ be the corresponding $n \times k$ data matrix and let $\mathbf{C}_Y$ be the $n \times q$ matrix whose column vectors are the first q principal components of $\mathbf{Y}$. Geometrically, the rows of $\mathbf{C}_Y$ can be thought of as an n-point configuration in q-dimensional space which approximates the 'true' configuration associated to $\mathbf{C}_X$. Obviously, the approximation is due to the fact that $\mathbf{C}_Y$ is based on $k < p$ variables. To evaluate the discrepancy between the configurations of $\mathbf{C}_X$ and $\mathbf{C}_Y$, Krzanowski suggests the use of Procrustes T^2-statistics, which is equal to the minimum value attainable by the sum of the squared distances between the corresponding points of the two configurations when one of these, e.g. $\mathbf{C}_Y$, is matched to the other by translations, rotations and reflections.

The optimal translation is that which superimposes the centroids of the two configurations.

Such a condition automatically holds when considering mean-centred data matrices. Consider now the optimal rotation/reflection. The SVD of the matrix $\mathbf{C}_X^T\mathbf{C}_Y$ $(q \times q)$ is

$$\mathbf{C}_X^T\mathbf{C}_Y = \mathbf{FRG}^T$$

where $\mathbf{F}$ and $\mathbf{G}$ are orthogonal matrices and $\mathbf{R} = \mathrm{diag}(r_1, \ldots, r_q)$, with $r_1 \geqslant \ldots \geqslant r_q \geqslant 0$. From the main results of Procrustes analysis, the optimal matching is obtained when the configuration of $\mathbf{C}_Y$ is transformed by the rotation/reflection defined by the orthogonal matrix $\mathbf{M} = \mathbf{FG}^T$. The T^2-statistics has the following expression:

$$T^2 = \mathrm{tr}(\mathbf{C}_X\mathbf{C}_X^T + \mathbf{C}_Y\mathbf{C}_Y^T - 2\mathbf{C}_Y^T\mathbf{C}_X\mathbf{M}) = \mathrm{tr}(\mathbf{C}_X\mathbf{C}_X^T + \mathbf{C}_Y\mathbf{C}_Y^T - 2\mathbf{R}) \tag{4}$$

Using equation (4), configurations $\mathbf{C}_Y$ corresponding to different choices of k-variable subsets can be compared with the reference configuration $\mathbf{C}_X$. The optimal subset has the minimum T^2-value of all possible subsets of k observed variables. Practical implementations of this method can be based on suboptimal stepwise algorithms of forward selection or backward elimination of variables. A simple and efficient procedure of backward elimination is described by Krzanowski.[7]

RESULTS AND DISCUSSION

Polyphenol data

Polyphenol substances predominantly contribute to the organoleptic characteristics of red wines, mainly the taste, the colour and, in part, the aroma.

A number of measurements concerning the content of polyphenol substances were made on 22 samples of three celebrated Italian red wines, Chianti Classico (five samples, 1982 vintage), Brunello di Montalcino (five samples, 1982 vintage) and Marzemino del Trentino (twelve samples, 1976 vintage). Chianti and Brunello wines are produced in contiguous areas in Tuscany (central Italy); Marzemino wine is produced in the Trentino region (north Italy). The Chianti wine is obtained from mixed grapes with a 50% minimum of the 'Sangiovese' variety, while the Brunello wine is produced from the grape variety 'Sangiovese grosso'. Thus these two wines are expected not to be very different.

Samples were analysed to determine[20,21] total phenols (two different methods), polymerizable phenols, antocyanins and leucoantocyanins. The five different measurements were made (a) on untreated wines, (b) on wines treated with cinchonine sulphate (substances with tannic properties removed by precipitation), (c) on wines treated with sodium chloride (polymeric substances of between six and ten units removed by precipitation) and (d) on the extract obtained with ethyl acetate on NaCl-treated wines (only polyphenols with a low degree of polymerization or monomer are extracted). Overall means, minimum and maximum values for each type of wine and the between-groups variance to total variance ratios are given in Table 1 together with a brief description of the observed variables.

Preliminary analyses showed two main features of the data. A number of variables have notably different distributions for the three types of wine. In particular, as shown by the variance ratios in the fourth column of Table 1, almost all of the variability of the leucoantocyanins (E-variables) arises from the differences between the means of the three wines. Moreover, pairwise linear correlations (see Table 2) are generally high, e.g. $\mathrm{corr}(A1, B1) = \mathrm{corr}(E1, E3) = 0{\cdot}99$, $\mathrm{corr}(A3, B3) = 0{\cdot}97$ and $\mathrm{corr}(A4, B4) = 0{\cdot}96$. However, the polymerizable phenols (C-variables), excluding C1, have comparatively low correlations with the other variables.

Table 1. Polyphenol data. Summary statistics

Chemical variable[a]	Overall mean $(mg\,l^{-1}) \times 10^{-2}$	Minimum–maximum $(mg\,l^{-1}) \times 10^{-2}$			Variance ratio[b] (%)
		Chianti	Brunello	Marzemino	
A1	29·8	28·2–32·4	35·3–45·7	18·8–41·4	61
B1	17·7	16·4–18·8	20·0–29·6	10·4–24·8	63
C1	4·7	4·2–5·0	4·8–8·1	2·4–8·4	31
D1	4·1	4·2–5·6	6·6–7·9	1·0–7·4	66
E1	19·5	24·2–31·0	29·7–51·6	3·2–14·1	93
B2	9·6	9·1–11·1	10·4–12·3	6·7–12·5	39
C2	2·3	2·1–2·4	1·8–2·9	1·3–3·6	<1
D2	2·7	2·9–4·6	3·2–5·1	0·5–4·3	60
E2	6·5	9·6–12·1	9·4–12·6	1·7–5·4	95
A3	20·7	19·5–23·1	20·3–26·6	15·9–24·8	39
B3	12·2	11·3–13·3	12·0–16·2	8·4–15·2	40
C3	3·2	3·2–3·3	2·6–4·0	1·8–5·4	<1
D3	2·0	1·8–2·7	2·3–3·0	0·8–3·4	34
E3	10·9	13·2–15·5	17·5–25·3	2·6–7·8	95
A4	9·8	10·3–12·2	8·2–13·3	5·8–11·3	41
B4	5·5	6·0–6·3	4·7–7·5	2·9–7·3	26
C4	2·1	1·9–2·2	1·4–2·8	1·1–3·6	<1
E4	3·6	5·8–7·2	3·4–6·0	1·1–3·1	90

[a] Code:
A total phenols measured at 280 nm (as (+)catechin)
B total phenols by Folin–Ciocalteau method (as (+)catechin)
C polymerizable phenols reactive to vanillin (as (+)catechin)
D antocyanins (as cyanidin)
E leucoantocyanins (as cyanidin)
1 untreated wine
2 after addition of cinchonine sulphate
3 after addition of sodium chloride
4 on ethyl acetate extract after NaCl treatment.
(Example: A3, total phenols measured at 280 nm after addition of sodium chloride.)
[b] Between-groups variance to total variance.

On the grounds of these results, two main questions were asked. First, how many factors are needed in the bilinear model to 'capture' the main features of the data? Secondly, is it possible to reduce the number of observed variables by finding a subset of them still retaining the valuable information about the groups?

Since all the variables have the same measurement units and their order of magnitude is comparable, the bilinear model was fitted to the unstandardized observed data. The first five eigenvalues of the covariance matrix, the corresponding jack-knife estimates and the W-statistics are given in Table 3. Clearly the first component has a dominant role: the first eigenvalue by itself equals 92% of the total variation. Moreover, the sum of the first two eigenvalues gives almost all of the total variation (about 98%). From Table 4 it can be seen that the first component is mostly correlated with leucoantocyanins (E-variables) and total phenols (A- and B-variables) while the second has greatest correlations with the polymerizable phenols (C-variables). The W-statistics shows values higher than unity on the first two components, is equal to 0·73 and 0·84 for the third and fourth component respectively and thereafter is never greater than 0·3.

Table 2. Polyphenol data. Correlation matrix

	A1	B1	C1	D1	E1	B2	C2	D2	E2	A3	B3	C3	D3	E3	A4	B4	C4	E4
A1	1																	
B1	0·99	1																
C1	0·92	0·93	1															
D1	0·92	0·90	0·79	1														
E1	0·85	0·87	0·65	0·83	1													
B2	0·93	0·91	0·91	0·88	0·72	1												
C2	0·51	0·48	0·69	0·40	0·09	0·69	1											
D2	0·83	0·78	0·65	0·95	0·77	0·83	0·39	1										
E2	0·75	0·75	0·54	0·79	0·94	0·70	0·10	0·79	1									
A3	0·93	0·93	0·92	0·80	0·74	0·94	0·64	0·71	0·67	1								
B3	0·91	0·93	0·94	0·76	0·73	0·91	0·59	0·63	0·64	0·97	1							
C3	0·61	0·60	0·82	0·45	0·23	0·75	0·89	0·36	0·21	0·74	0·75	1						
D3	0·78	0·74	0·68	0·92	0·58	0·83	0·55	0·91	0·55	0·70	0·63	0·49	1					
E3	0·85	0·87	0·65	0·84	0·99	0·71	0·08	0·76	0·93	0·74	0·74	0·22	0·58	1				
A4	0·83	0·83	0·81	0·73	0·74	0·88	0·62	0·71	0·75	0·91	0·86	0·72	0·63	0·71	1			
B4	0·83	0·82	0·88	0·73	0·63	0·90	0·76	0·66	0·62	0·91	0·89	0·85	0·68	0·62	0·96	1		
C4	0·59	0·58	0·78	0·46	0·21	0·71	0·95	0·41	0·18	0·70	0·67	0·95	0·54	0·19	0·71	0·84	1	
E4	0·65	0·65	0·49	0·66	0·85	0·60	0·13	0·68	0·93	0·58	0·56	0·29	0·43	0·81	0·76	0·65	0·27	1

Table 3. Polyphenol data. Main results from cross-validatory estimation of the bilinear model: eigenvalues (l_j), jack-knife estimates of the eigenvalues ($\hat{l}_j$), jack-knife estimates of the standard deviations of $\hat{l}_j$ ($\hat{s}_j$) and W-statistics

Component no.	$l_j^{(a)}$ $\times 10^{-6}$	$\hat{l}_j$ $\times 10^{-6}$	$\hat{s}_j$ $\times 10^{-6}$	W_j
1	4·34	4·53	0·98	45·73
2	0·29	0·31	0·13	9·03
3	0·04	0·04	0·01	0.73
4	0·03	0·03	0·01	0·84
5	0·01	0·01	<0·01	0·30

[a] Total variation $4{\cdot}72 \times 10^6$

Table 4. Polyphenol data. Correlations of the observed variables with the first two principal components

Chemical variable	Principal component 1st	2nd
A1	0·92	0·40
B1	0·93	0·36
C1	0·75	0·63
D1	0·88	0·26
E1	0·99	−0·14
B2	0·80	0·52
C2	0·21	0·85
D2	0·81	0·17
E2	0·92	−0·22
A3	0·82	0·50
B3	0·81	0·49
C3	0·35	0·81
D3	0·65	0·44
E3	0·99	−0·13
A4	0·79	0·33
B4	0·71	0·51
C4	0·32	0·81
E4	0·83	−0·24

These results suggest that the bilinear model with the first two principal components should provide an adequate fit to the observed data. In fact, the squared Euclidean norms of the estimated residuals (see the index plot in Figure 1(a)) reveal no important differences between observed and predicted values nor any systematic pattern of the values, even though there is a tendency for Chianti and Brunello samples to have the largest residuals. It is interesting to contrast this plot with the rather different index plot derived from the *predicted* residuals (see Figure 1(b)) in which two samples (8 and 15) have Q-values much larger than those of the other samples. This fact denotes that these two samples have very poor predictions. Also, the scatter

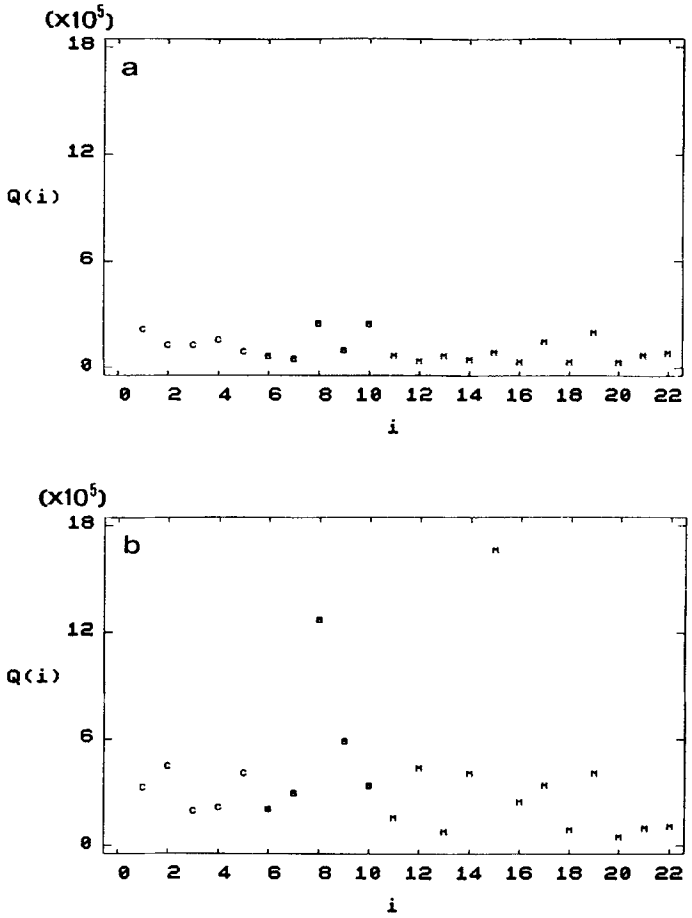

Figure 1. Polyphenol data. Plot of $Q(i)$ in serial order: (a) estimated residuals; (b) predicted residuals. C, Chianti; B, Brunello; M, Marzemino

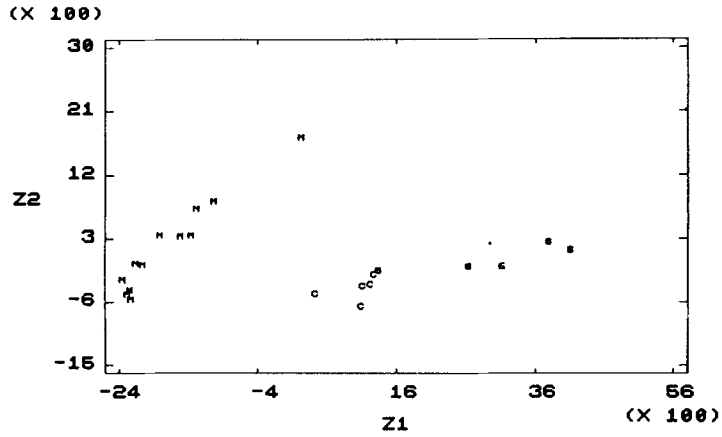

Figure 2. Polyphenol data. Scatter plot of the first two principal components Z1 and Z2

diagram of the observations in the plane of the first two principal components (Figure 2) shows three clusters clearly corresponding to the three types of wine.

The SICs of the first two eigenvalues (see Figure 3) show three influential samples of Brunello wine on the first eigenvalue (samples 7–9) and one highly influential sample of Marzemino wine on the second eigenvalue (sample 15). Similar remarks apply to $(n-1)\|\tilde{\mathbf{u}}_j - \tilde{\mathbf{u}}_{(i)j}\|$ and $\|\mathbf{C}_j - \mathbf{C}_{(i)j}\|$, $j=1,2$: in fact, $\mathrm{corr}(\mathrm{SIC}_i(\lambda_1), (n-1)\|\tilde{\mathbf{u}}_1 - \tilde{\mathbf{u}}_{(i)1}\|) = 0{\cdot}78$, $\mathrm{corr}(\mathrm{SIC}_i(\lambda_2), (n-1)\|\tilde{\mathbf{u}}_2 - \tilde{\mathbf{u}}_{(i)2}\|) = 0{\cdot}97$ and $\mathrm{corr}(\mathrm{SIC}_i(\lambda_1), \|\mathbf{C}_1 - \mathbf{C}_{(i)1}\|) = 0{\cdot}96$. Now, the samples of Brunello wine have polyphenol concentrations remarkably higher than those of Marzemino and Chianti wines, particularly for the variables more correlated with the first principal component, i.e. E1 and E3 (corr = 0·99), B1 (corr = 0·93) and A1 and E2 (corr = 0·92). Thus the exclusion of Brunello samples determines a strong decrease in the variance (i.e. the eigenvalue) of the first component. A possible explanation is that the region of the sample space with higher polyphenol concentrations is insufficiently represented in the data set and thus four out of five samples of Brunello wine appear very dissimilar with respect to the other wines. This is not the case in the interpretation of the Marzemino sample. Inspection of the data showed untypical values for almost all the observed variables, so this sample was discarded from the data set as an outlier.

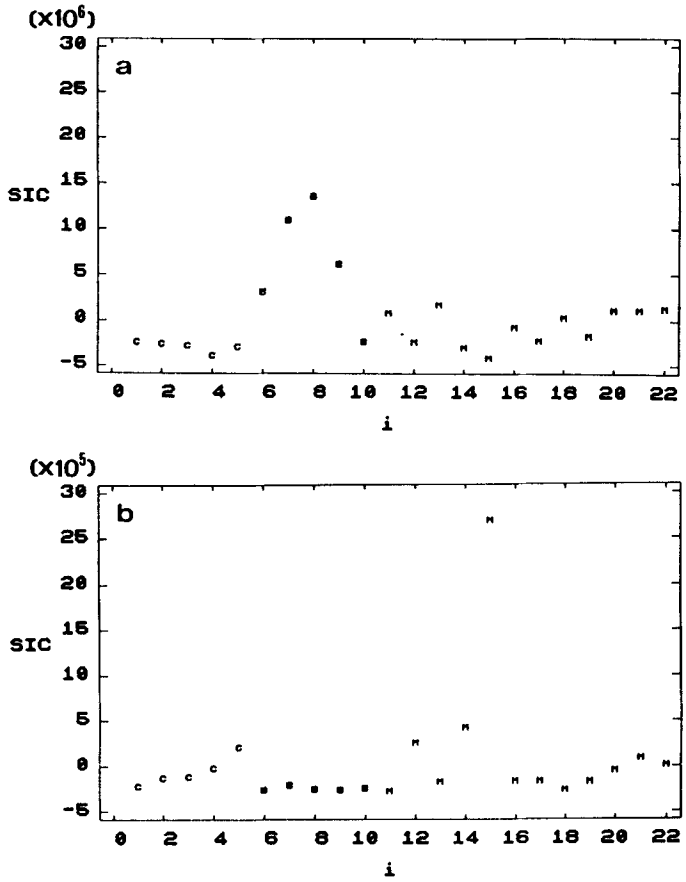

Figure 3. Polyphenol data. Plot of $\mathrm{SIC}_i(\lambda_j)$ in serial order: (a) $j=1$; (b) $j=2$

Table 5. Polyphenol data (sample 15 discarded). Main results from cross-validatory estimation of the bilinear model. Symbol meanings as in Table 3

Component no.	$l_j^{(a)}$ $\times 10^{-6}$	$\hat{l}_j$ $\times 10^{-6}$	$\hat{s}_j$ $\times 10^{-6}$	W_j
1	4·54	4·76	1·01	68·51
2	0·17	0·17	0·04	5·48
3	0·04	0·05	0·01	1·16
4	0·02	0·02	0·01	0·72
5	0·01	0·02	<0·01	0·46

[a] Total variation $4{\cdot}80 \times 10^6$

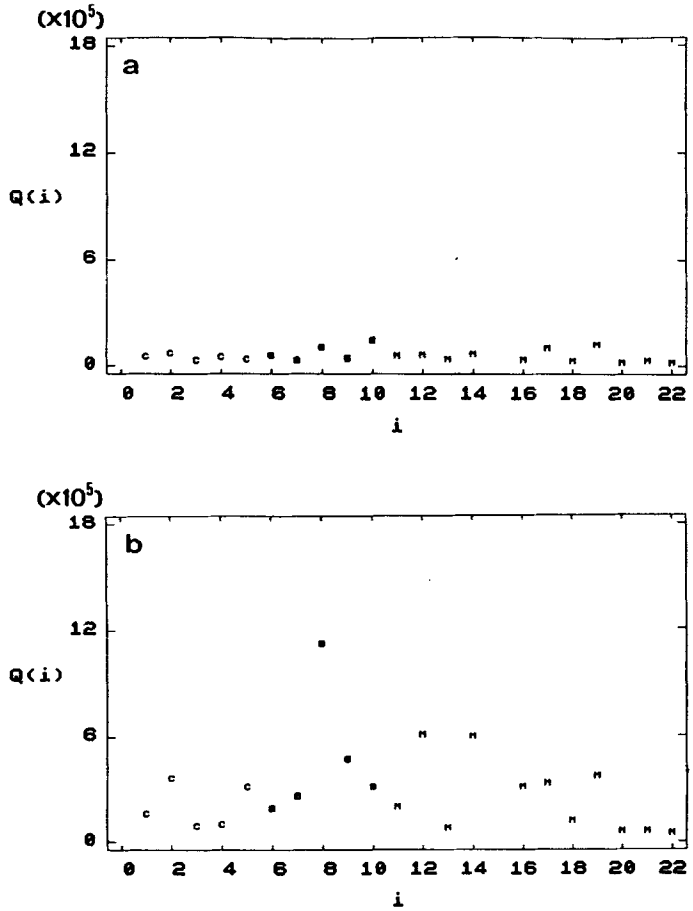

Figure 4. Polyphenol data (sample 15 discarded). Plot of $Q(i)$ in serial order: (a) estimated residuals; (b) predicted residuals

PCA results obtained after the outlying Marzemino sample was discarded are given in Table 5. From the W-statistics the number of 'significant' components is now equal to three, while on the complete data set it turned out equal to two. The first three components account for 99% of the total variation, so the overall fitting of the model is very good. Also, from the Q-values, with the exception of sample 8, there is no evidence of important residuals (see Figure 4). The correlation pattern of the first two principal components with the observed variables is almost unchanged and the third component has major correlations with the variables measured on ethyl acetate extract and with C2 and C3 (see Table 6).

The procedure of backward elimination of variables based on Krzanowski's algorithm retained the three-variable subset {A1, E1, E3} whose T^2-value is $2 \cdot 04 \times 10^6$. When the procedure was tried with $k = 4$ and $k = 5$, E2 and then B1 were added to the previous subset and T^2 ({A1, B1, E1, E2, E3}) $= 0 \cdot 67 \times 10^6$. Using Jolliffe's 'B2' and 'B4' methods,[4] the selected three-variable subsets are {A1, B1, E1} and {E1, C2, A4} respectively. The Procrustes statistics corresponding to these subsets are T^2({A1, B1, E1}) $= 2 \cdot 52 \times 10^6$ and T^2({E1, C2, A4}) $= 8 \cdot 37 \times 10^6$. With $k = 5$, 'B2' produces the same subset obtained through Krzanowski's method while 'B4' gives the subset {E1, C2, D3, A4, C4} whose T^2-value is very far from the (sub)optimal one ($T^2 = 8 \cdot 25 \times 10^6$).

The scatter diagram of the first three principal components of the five-variable subset {A1, B1, E1, E2, E3} matches very well with the scatter diagram based on all the variables (see Figure 5); in particular, the three-group clustering and the shape of the groups are confirmed. Thus, in this example, the number of observed variables can be substantially reduced while still preserving the bulk of the information supplied by the data.

Table 6. Polyphenol data (sample 15 discarded). Correlations of the observed variables with the first three principal components

Chemical variable	Principal component		
	1st	2nd	3rd
A1	0·95	0·28	−0·04
B1	0·96	0·28	−0·07
C1	0·84	0·49	0·03
D1	0·92	0·07	−0·04
E1	0·99	−0·09	−0·01
B2	0·86	0·39	0·22
C2	0·23	0·79	0·44
D2	0·83	0·01	0·12
E2	0·93	−0·23	0·26
A3	0·85	0·45	0·17
B3	0·84	0·44	0·11
C3	0·41	0·70	0·46
D3	0·71	0·22	0·02
E3	0·99	−0·08	−0·05
A4	0·80	0·33	0·48
B4	0·74	0·45	0·44
C4	0·36	0·72	0·42
E4	0·83	−0·27	0·43

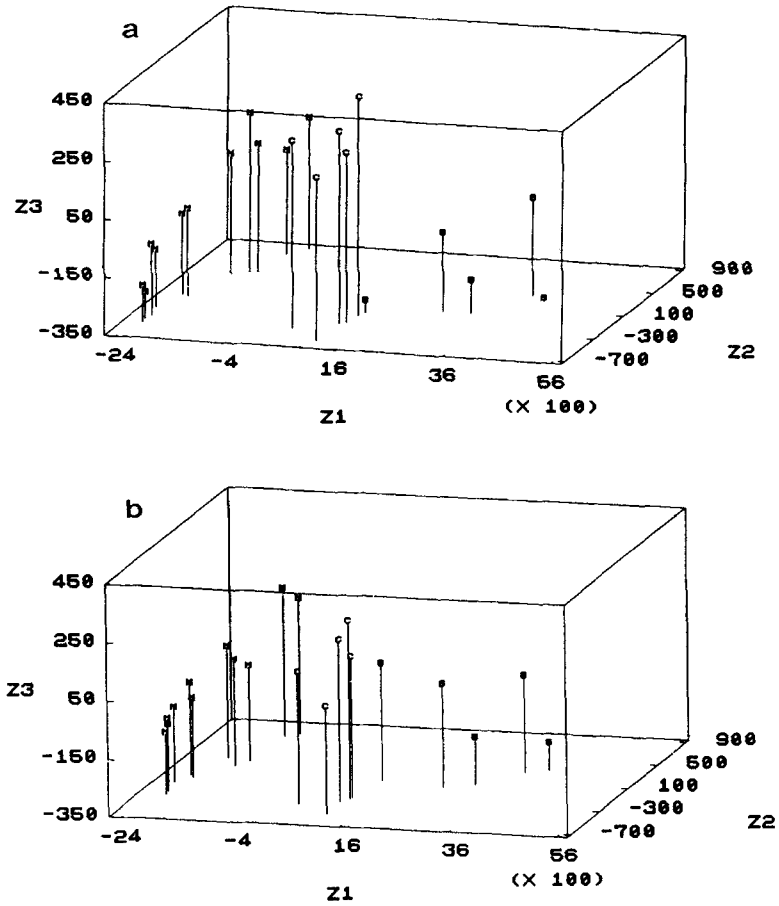

Figure 5. Polyphenol data (sample 15 discarded). Scatter plot of the first three principal components Z1, Z2 and Z3 derived from (a) all the variables and (b) the subset {A1, B1, E1, E2, E3}

Prosecco data

This data set includes observations on 34 samples of the Prosecco white wine (produced in the Veneto region of north-east Italy) from the 1982, 1983 and 1984 vintages. Nineteen classical chemical variables and the concentrations of nine aroma compounds were measured on each sample. Details of the chemical procedures are given elsewhere.[22] The list of the variables together with summary statistics are given in Table 7.

In an earlier work[22] (where samples from 1982 and 1983 vintages only were considered) the familiar percentage trace criterion suggested that at least the first four or five components are needed to explain 75%–80% of the total variation. Now cross-validatory estimation of the bilinear model should allow a more formal individuation of the components with actual predictive ability.

After a preliminary standardization of the observed variables to unit standard deviations, the fitting of the bilinear model gave the results which are shown in Table 8. Using Krzanowski's criterion, the first eight components are significant. However, two rather peculiar facts are to

Table 7. Prosecco data. Summary statistics

Chemical variable[a]	Measurement unit	Mean	Minimum–maximum	Standard deviation	Coefficient of variation (%)
NA	$mg\,l^{-1} \times 10^{-1}$	1·5	0·3–3·7	0·8	51
K	$mg\,l^{-1} \times 10^{-2}$	5·9	2·8–8·1	1·2	21
CA	$mg\,l^{-1} \times 10^{-1}$	9·8	7·9–13·8	1·4	14
MG	$mg\,l^{-1} \times 10^{-1}$	7·1	5·4–9·1	0·8	11
CL	$mg\,l^{-1} \times 10^{-1}$	1·8	0·3–3·5	0·7	40
PH		3·27	2·94–3·59	0·15	4
TA	$g\,l^{-1}$	5·8	4·5–7·7	0·9	16
P	$mg\,l^{-1} \times 10^{-2}$	1·5	0·6–3·0	0·6	38
AC	$g\,l^{-1}$	1·6	1·0–2·1	0·3	17
ALA	$meq\,l^{-1} \times 10^{-1}$	1·9	1·2–2·3	0·3	14
LI	$\mu g\,l^{-1} \times 10^{-1}$	1·1	0·8–1·6	0·2	17
TN	$mg\,l^{-1} \times 10^{-2}$	1·4	0·7–3·2	0·5	34
DE	$g\,l^{-1} \times 10^{-1}$	1·8	1·4–2·6	0·3	15
A	$\times 10$	1·3	0·5–5·8	1·0	73
TP	$mg\,l^{-1} \times 10^{-2}$	2·8	1·4–13·0	1·9	66
POL	$mg\,l^{-1} \times 10^{-1}$	2·9	1·7–4·5	0·7	24
MPOL	$mg\,l^{-1} \times 10^{-1}$	6·5	3·0–11·5	2·1	32
MBOL	$mg\,l^{-1} \times 10^{-2}$	2·2	0·9–3·4	0·5	25
HOLS	$mg\,l^{-1} \times 10^{-2}$	3·1	1·7–4·8	0·7	21
MPA	$mg\,l^{-1}$	1·3	0·5–2·3	0·4	30
BA	$mg\,l^{-1}$	1·1	0·4–2·8	0·5	44
MBA	$mg\,l^{-1}$	0·7	0·3–1·4	0·2	32
HA	$mg\,l^{-1}$	2·9	1·3–5·3	1·0	33
OA	$mg\,l^{-1}$	4·8	1·9–10·2	1·8	37
DA	$mg\,l^{-1}$	1·3	0·3–3·8	0·7	53
HOL	$mg\,l^{-1}$	2·2	0·9–3·8	0·8	36
THOL	$mg\,l^{-1} \times 10$	0·6	0·2–1·2	0·3	51
CHOL	$mg\,l^{-1} \times 10$	1·1	0·4–2·2	0·4	32

[a] Code: NA sodium, K potassium, CA calcium, MG magnesium, CL chloride, PH pH, TA titratable acidity as tartaric acid, P phosphorus as P_2O_5, AC ash content, ALA alkalinity of the ash, LI lithium, TN total nitrogen, DE theoretical sugar-free dry extract, A absorbance measured at 420 nm, TP total phenols as (+)catechin, POL 1-propanol, MPOL 2-methyl-1-propanol, MBOL 2-methyl-1-butanol plus 3-methyl-1-butanol, HOLS (=POL+MPOL+MBOL) higher alcohols, MPA 2-methyl-propanoic acid, BA butanoic acid, MBA 3-methyl-butanoic acid, HA hexanoic acid, OA octanoic acid, DA decanoic acid, HOL 1-hexanol, THOL *trans*-3-hexen-1-ol, CHOL *cis*-3-hexen-1-ol.

be noted. The values of the W-statistics are far from monotonic decreasing and the second and fifth components have *negative* W-values.

Non-monotonicity can have the following explanation.[15] When two eigenvalues of the population covariance (or correlation) matrix are not well separated, it can happen that the ordering of the *sample* principal components does not match the ordering of the *population* principal components. Thus, in the present case, as suggested by the jack-knifed eigenvalues and the corresponding jack-knife estimates of their standard deviations, the third sample principal component could be an estimate of the second population principal component. To overcome this difficulty, several orderings of the sample principal components should be tried to achieve correct estimation of the number of 'significant' components from the W-statistics.

Negative values of W deserve special consideration since they mean that the corresponding

Table 8. Prosecco data. Main results from cross-validatory estimation of the bilinear model. Symbol meanings as in Table 3

Component no.	$l_j^{(a)}$	$\hat{l}_j$	$\hat{s}_j$	$W_j^{(b)}$
1	6·94(25)	6·16	0·93	2·0
2	3·61(13)	2·52	0·45	−0·7
3	3·13(11)	2·71	0·68	1·0
4	2·58(9)	2·51	0·63	0·8
5	1·98(7)	0·67	0·11	−0·5
6	1·91(7)	2·40	0·43	0·9
7	1·56(6)	1·73	0·40	0·4
8	1·34(5)	2·36	0·46	0·9
9	0·93(3)	0·90	0·16	0·2
10	0·78(3)	0·90	0·16	0·2

[a] In parentheses: percentage of the total variation.
[b] No W_j is greater than 0.5 for $j > 10$.

components, when entered in the model, reduce the overall fitting as measured by the sum of the squared predicted residuals. In our experience this situation is more likely to occur when influential observations are present on the relevant eigenvalues and/or eigenvectors of the covariance (or correlation) matrix. In particular, influential observations can heavily affect eigenvectors associated to very close eigenvalues.[17]

The squared Euclidean norms of $\hat{\mathbf{e}}_i^{(1)}$ and $\hat{\mathbf{e}}_i^{(2)}$, $i = 1, \ldots, 34$ (see Figure 6), show that a number of samples (e.g. samples 10, 24 and 32) have very large predicted residuals on the first component. Also, it can be seen that the second component improves the fit for only eleven observations out of 34. Now, sample 10 has an abnormal (from the enological point of view) concentration of TP, while the concentrations of HA, OA and DA for sample 24 are twice as much as for the other samples, and sample 32 has a very low content of POL, MPOL, MBOL and HOLS.

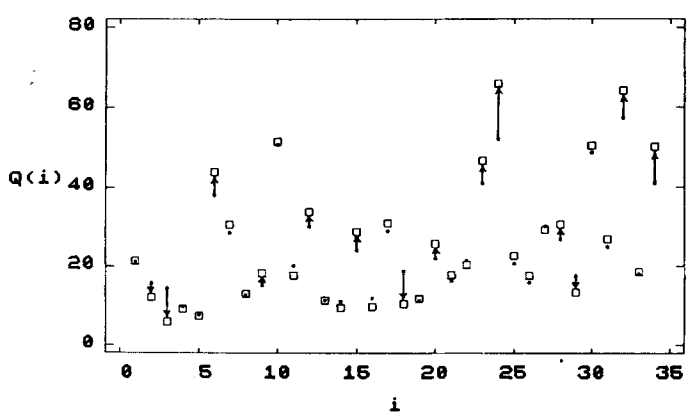

Figure 6. Prosecco data. Plot of the Euclidean squared norms of the predicted residuals in serial order: □, $Q(i) = \| \hat{e}_i^{(1)} \|^2$; ■, $Q(i) = \| \hat{e}_i^{(2)} \|^2$

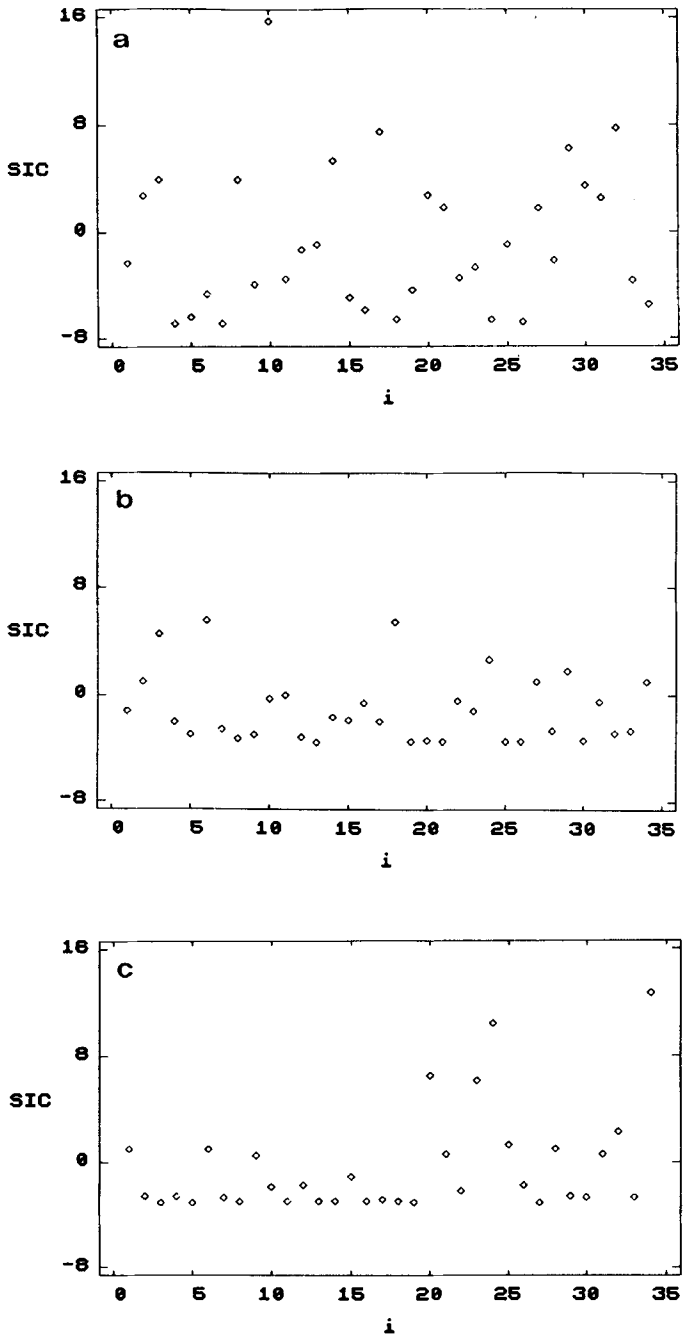

Figure 7. Prosecco data. Plot of $SIC_i(\lambda_j)$ in serial order: (a) $j = 1$; (b) $j = 2$; (c) $j = 3$

The SICs of the first three eigenvalues (Figure 7) show that sample 10 is influential on the first eigenvalue, while sample 24 and 34 are influential on the third. From Table 9(a), the loading of the first component on TP is nearly annihilated when the former sample is omitted from the data.

The diagrams of $P_i = (n-1)\,\|\,\tilde{\mathbf{u}}_j - \tilde{\mathbf{u}}_{(i)j}\,\|$, $j = 1, 2, 3$, given in Figure 8 show that the above-mentioned samples 10 and 32 are influential on $\mathbf{u}_1$ and that samples 6, 24 and 34 highly influence

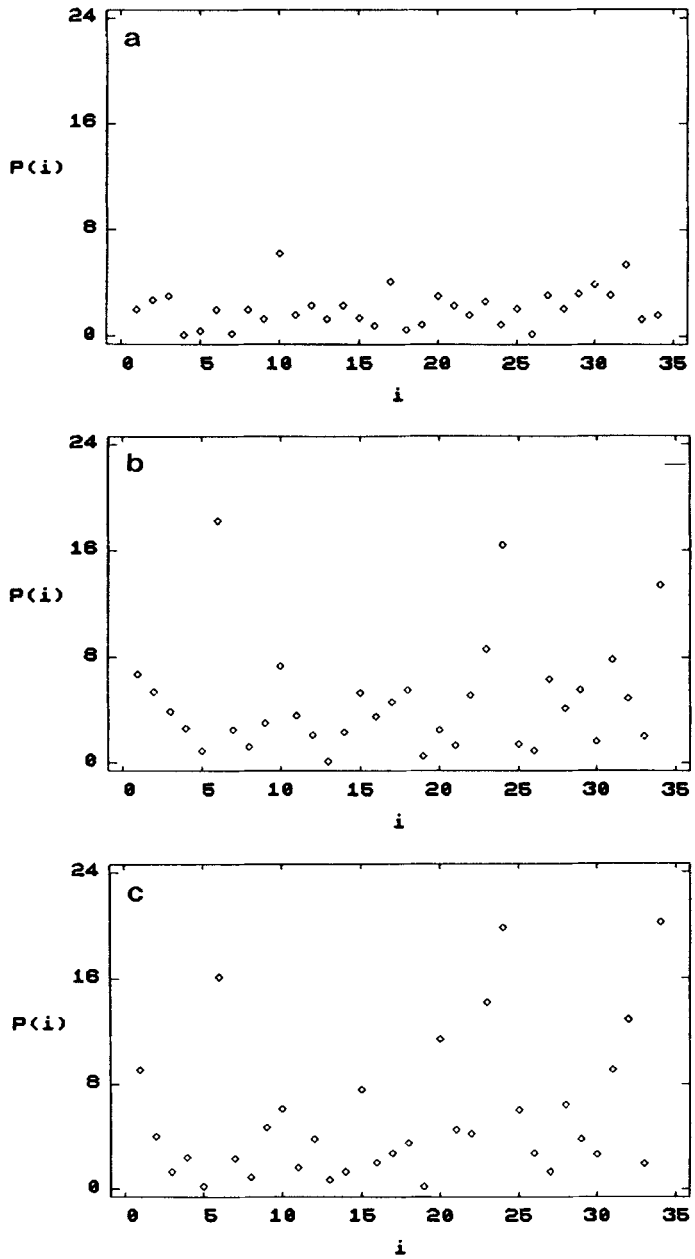

Figure 8. Prosecco data. Plot of $P(i) = (n-1)\,\|\,\tilde{\mathbf{u}}_j - \tilde{\mathbf{u}}_{(i)j}\,\|$ in serial order: (a) $j = 1$; (b) $j = 2$; (c) $j = 3$

Table 9. Prosecco data. The eigenvectors $\mathbf{u}_j$, $j = 1, 2$, of the correlation matrix and their perturbations $\mathbf{u}_{(i)j}$, $i = 6, 10, 24, 34$
(a) $j = 1$

Chemical variable	$\mathbf{u}_1$	$\mathbf{u}_{(6)1}$	$\mathbf{u}_{(10)1}$	$\mathbf{u}_{(24)1}$	$\mathbf{u}_{(32)1}$	$\mathbf{u}_{(34)1}$
NA	0·16	0·17	0·16	0·16	0·16	0·18
K	−0·23	−0·21	−0·21	−0·23	−0·22	−0·23
CA	0·03	0·04	0·04	0·02	0·06	0·02
MG	−0·22	−0·22	−0·22	−0·22	−0·19	−0·23
CL	−0·05	−0·04	−0·03	−0·05	−0·01	−0·06
PH	−0·26	−0·24	−0·22	−0·26	−0·27	−0·26
TA	0·25	0·25	0·27	0·25	0·25	0·26
P	−0·25	−0·26	−0·25	−0·25	−0·28	−0·26
AC	−0·26	−0·24	−0·23	−0·27	−0·24	−0·27
ALA	−0·20	−0·18	−0·19	−0·21	−0·18	−0·20
LI	0·11	0·13	0·17	0·11	0·16	0·12
TN	0·10	0·11	0·07	0·10	0·02	0·10
DE	0·08	0·08	0·08	0·08	0·11	0·08
A	<0·01	<0·01	0·07	<0·01	0·02	−0·01
TP	−0·17	−0·17	−0·05	−0·17	−0·18	−0·18
POL	−0·08	−0·08	−0·09	−0·07	−0·06	−0·08
MPOL	−0·04	−0·04	−0·05	−0·04	−0·04	−0·02
MBOL	−0·29	−0·29	−0·29	−0·29	−0·25	−0·28
HOLS	−0·26	−0·26	−0·27	−0·26	−0·23	−0·24
MPA	−0·04	−0·05	−0·04	−0·04	−0·02	−0·02
BA	0·19	0·21	0·23	0·19	0·19	0·19
MBA	−0·26	−0·27	−0·22	−0·27	−0·25	−0·25
HA	0·28	0·30	0·31	0·27	0·28	0·28
OA	0·26	0·28	0·28	0·25	0·29	0·27
DA	0·19	0·20	0·19	0·18	0·24	0·19
HOL	−0·23	−0·23	−0·28	−0·24	−0·27	−0·23
THOL	−0·05	−0·04	−0·07	−0·05	−0·04	−0·04
CHOL	0·07	0·08	0·05	0·07	<0·01	0·06

$\mathbf{u}_2$ and $\mathbf{u}_3$. From Table 9 it can be seen how much the loadings of the first two principal components are perturbed by these samples.

Previous results demonstrate that in the data set there are several highly influential observations which could be at the root of both non-monotonicity and negative values of W. Among these observations, samples 10, 24 and 32 appear to be somewhat untypical from the enological viewpoint, so the entire analysis was repeated omitting these 'suspect' samples and trying different orderings of the sample principal components.

When the principal components are entered in the bilinear model following the decreasing ordering of the sample eigenvalues, the number of 'significant' components is equal to seven, but the fourth and sixth components still have negative W-values (see Table 10). However, when the third and seventh components are entered in the model just after the first, the W-ratio is equal to 2·49 on the first component, is about unity on the third and seventh components and then is never greater than 0·6. Thus there is some evidence that the first component is 'significant', the third and seventh are still important, while the second gives very little improvement to the predictive capabilities of the model.

From Figure 9, the squared Euclidean norms of the predicted residuals $\hat{\mathbf{e}}_i^{(1)}$ and $\hat{\mathbf{e}}_i^{(1,3,7)}$ have

Table 9. continued
(b) $j = 2$

Chemical variable	$\mathbf{u}_2$	$\mathbf{u}_{(6)2}$	$\mathbf{u}_{(10)2}$	$\mathbf{u}_{(24)2}$	$\mathbf{u}_{(32)2}$	$\mathbf{u}_{(34)2}$
NA	−0·11	−0·27	−0·14	−0·19	−0·13	0·02
K	0·26	0·10	0·27	0·18	0·26	0·31
CA	0·23	0·11	0·23	0·12	0·21	0·27
MG	0·16	0·24	0·20	0·18	0·17	0·10
CL	0·27	0·31	0·27	0·27	0·25	0·19
PH	0·25	0·18	0·25	0·20	0·26	0·24
TA	−0·06	−0·13	−0·11	−0·08	−0·08	−0·01
P	0·01	0·14	0·05	0·12	0·05	−0·09
AC	0·32	0·20	0·32	0·23	0·31	0·32
ALA	0·23	0·06	0·24	0·12	0·22	0·28
LI	0·26	0·12	0·18	0·11	0·21	0·35
TN	−0·07	−0·06	−0·03	0·03	−0·02	−0·13
DE	0·02	0·02	0·01	0·03	0·01	0·02
A	0·14	0·13	0·07	0·15	0·12	0·13
TP	0·16	0·15	0·08	0·15	0·17	0·15
POL	−0·10	−0·17	−0·08	−0·10	−0·11	−0·08
MPOL	−0·28	−0·46	−0·31	−0·40	−0·31	−0·10
MBOL	−0·09	−0·16	−0·10	−0·22	−0·12	0·03
HOLS	−0·17	−0·30	−0·19	−0·32	−0·21	−0·02
MPA	−0·32	−0·42	−0·38	−0·46	−0·36	−0·14
BA	0·19	0·06	0·14	0·09	0·17	0·25
MBA	−0·06	−0·08	−0·11	−0·15	−0·08	0·01
HA	0·24	0·10	0·20	0·12	0·22	0·31
OA	0·21	0·07	0·17	0·05	0·17	0·29
DA	0·16	0·06	0·13	−0·01	0·12	0·24
HOL	−0·08	−0·14	−0·01	−0·10	−0·05	−0·06
THOL	0·11	0·06	0·15	0·06	0·11	0·13
CHOL	0·13	0·11	0·17	0·19	0·18	0·08

Table 10. Prosecco data (samples 10, 24 and 32 discarded). Main results from cross-validatory estimation of the bilinear model. Symbol meanings as in Table 3

Component no.	$l_j^{(a)}$	$\hat{l}_j$	$\hat{s}_j$	$W_j^{(b)}$	Component no.	$W_j^{(b)}$
1	7·43(27)	6·73	1·03	2·49	1	2·49
2	4·07(15)	3·25	0·66	0·30	3	1·04
3	3·08(11)	2·95	0·78	1·12	7	1·06
4	2·29(8)	1·57	0·26	−0·29	8	0·55
5	2·06(7)	2·06	0·39	0·55	5	0·60
6	1·67(6)	0·92	0·19	−0·04	2	0·52
7	1·53(5)	2·23	0·38	1·17	6	0·11
8	1·20(4)	1·28	0·33	0·66	4	−0·13
9	0·96(3)	1·43	0·34	0·29	9	0·29
10	0·72(3)	0·58	0·12	0·11	10	0·11

[a] In parentheses: percentage of the total variation.
[b] No W_j is greater than 0·40 for $j > 10$.

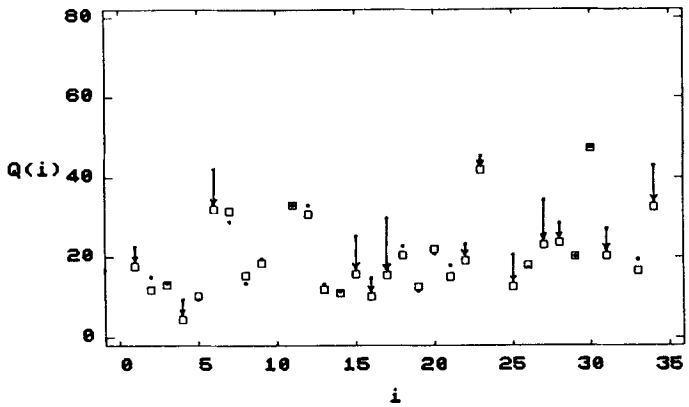

Figure 9. Prosecco data (samples 10, 24, 32 discarded). Plot of the Euclidean squared norms of the predicted residuals in serial order: ▪, $Q(i) = \| \hat{e}_i^{(1)} \|^2$; □, $Q(i) = \| \hat{e}_i^{(1,3,7)} \|^2$

Table 11. Proseco data (samples 10, 24 and 32 discarded). Linear correlation of selected principal components with the observed variables

Chemical variable	Principal component			
	1st	2nd	3rd	7th
NA	0·41	0·32	0·59	−0·10
K	−0·55	−0·44	0·63	−0·06
CA	0·13	−0·30	0·41	0·28
MG	−0·53	−0·32	−0·28	0·25
CL	<0·01	−0·46	−0·20	0·04
PH	−0·65	−0·41	0·17	−0·08
TA	0·65	0·15	0·28	0·35
P	−0·68	−0·19	−0·28	−0·09
AC	−0·64	−0·53	0·39	0·01
ALA	−0·46	−0·33	0·56	−0·31
LI	0·50	−0·22	0·25	0·35
TN	−0·04	−0·08	0·33	0·02
DE	0·27	−0·07	−0·08	0·41
A	0·23	−0·22	0·04	0·23
TP	−0·46	−0·52	−0·12	0·14
POL	−0·16	0·13	0·54	−0·09
MPOL	−0·07	0·64	0·61	−0·13
MBOL	−0·73	0·46	0·04	0·13
HOLS	−0·63	0·61	0·31	0·05
MPA	−0·03	0·85	0·15	0·25
BA	0·51	−0·18	0·33	0·46
MBA	−0·61	0·42	−0·16	0·45
HA	0·83	−0·28	0·30	−0·07
OA	0·84	−0·17	0·15	−0·13
DA	0·70	−0·05	−0·07	−0·17
HOL	−0·79	0·09	0·28	0·13
THOL	−0·20	−0·20	0·03	0·42
CHOL	−0·07	−0·54	0·30	0·13

a better pattern than in the previous case. We note that the first component has high linear correlations with HA (corr = 0·83), OA (corr = 0·84), HOL (corr = −0·79), MBOL (corr = −0·73) and P (corr = −0·68) and the third component has the highest correlations with NA, K, ALA and MPOL (corr ≃ 0·6). Other details are given in Table 11.

The procedure of selection of variables based on the Procrustes criterion was applied to the 'deleted' data set, taking as the reference configuration (a) the configuration of the first seven principal components and (b) the configuration of the first, third, fifth, seventh and eighth principal components. In the first case the (sub)optimal seven-variable subset is {ALA, DE, MPOL, MBOL, BA, OA, CHOL} while in the second it turns out to be {MG, ALA, TN, MPOL, MBOL, OA, CHOL}.

These results show that the characteristics of the Prosecco data are quite different from the Polyphenol data. Here it is suggested that several components should be entered in the model to describe the structure of the data adequately. More disturbingly, because of the non-monotonicity and the negative values of W, it is not quite clear whether the selected components are truly significant. On the other hand, the (sub)optimal Procrustean subsets are not 'random' but include variables which are responsible for the correlation structure of the data.[22]

CONCLUDING REMARKS

The bilinear model based on principal components accomplishes two main tasks: it deepens our knowledge of the structure of multivariate data and at the same time it can be used as a prediction tool. However, some difficulties accompany its use, above all the determination of the number of factors to be entered in the model. Cross-validatory estimation is no doubt better than methods of a more subjective nature such as the percentage trace criterion and the rule based on the 'least greater than one' eigenvalue (when working with standardized data). Nevertheless, statistics such as the W-ratio can be difficult to interpret because of negative values and non-monotonicity. Another problem is the absence of distributional results and thus of significance tests. To solve this latter problem in the general small-sample case, a distribution (or at least some properties involving the covariance matrix) for the random component of the model should be described. At present, in the standard least squares setting, this distribution is left completely undetermined.

Residual analysis and influence analysis deserve routine use, in the first case to check model fitting, in the second to ensure that the results are not dominated by a single observation. As shown in this work, these diagnostics can be of use in revealing gross errors in the data and extreme samples. The open questions include the properties of the different types of residuals, the definition of probabilistic or at least empirical criteria to settle when influence is large, and the generalization to *groups* of influential observations.

The screening of important variables is an old problem; now it is becoming important in chemometrics because of the increased use of semi-automatic instruments able to measure several tens of variables on the same sample. Procrustes analysis is a general method for comparing a test configuration with a reference configuration, and the previous examples showed that it provides sensible solutions to the variable selection problem. The function which summarizes the discrepancy between the configurations (the Procrustes statistics T^2) stresses the geometrical aspects of the matching, while the criteria suggested by Jolliffe and McCabe are based on the correlation structure of the variables. Thus we expect that they can select quite different subsets with respect to the Procrustes procedure. We hope shortly to give a comparison of these methods based on real data.

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REFERENCES

1. R. D. Cook and S. Weisberg, *Residuals and Influence in Regression*, Chapman and Hall, London (1982).
2. M. L. Thompson, *Int. Stat. Rev.* **46**, 1, 129 (1978).
3. R. J. McKay and N. A. Campbell, *Br. J. Math. Stat. Psychol.* **35**, 1, 30 (1982).
4. I. T. Jolliffe, *Appl. Stat.* **21**, 160 (1972).
5. I. T. Jolliffe, *Appl. Stat.* **22**, 21 (1973).
6. G. P. McCabe, *Technometrics*, **26**, 137 (1984).
7. W. J. Krzanowski, *Appl. Stat.* **36**, 22 (1987).
8. S. Wold, *Pattern Recognition*, **8**, 127 (1976).
9. K. R. Gabriel, *J. R. Stat. Soc. B*, **40**, 186 (1978).
10. K. V. Mardia, J. T. Kent and J. M. Bibby, *Multivariate Analysis*, Academic Press, London (1979).
11. S. Wold, *Technometrics*, **20**, 397 (1978).
12. H. T. Eastment and W. J. Krzanowski, *Technometrics*, **24**, 73 (1982).
13. J. R. Bunch and C. P. Nielsen, *Numer. Math.* **31**, 111 (1978).
14. J. R. Bunch, C. P. Nielsen and D. C. Sorensen, *Numer. Math.* **31**, 31 (1978).
15. W. J. Krzanowski, *Biometrics*, **43**, 575 (1987).
16. J. E. Jackson and G. S. Mudholkar, *Technometrics*, **21**, 341 (1979).
17. F. Critchley, *Biometrika*, **72**, 627 (1985).
18. I. T. Jolliffe, *Principal Component Analysis*, Chap. 10, Springer-Verlag, New York (1986).
19. R. G. Miller, *Biometrika*, **61**, 1 (1974).
20. G. Margheri, D. Tonon and C. Mattarei, *Vini d'Italia*, **19**, 301 (1977).
21. G. Margheri, I. Moret and G. Scarponi, personal communication.
22. I. Moret, G. Capodaglio, G. Scarponi and M. Romanazzi, *Anal. Chim. Acta*, **191**, 331 (1986).